CMake Project and Engine Management

Applies to FemboyEngine2D 0.4.0-alpha.1 and CMake 3.28 or newer.

Toolchain

The engine requires C++26:

- Windows: clang-cl 23.1.4+ (reference) or MSVC 19.44+ (optional alternative), both with the MSVC STL.
- Linux: Clang 23.1.4+ or GCC 16.2.0+.
- macOS: Apple Clang with C++26 support.

Ninja is the supported generator for the validated build. Visual Studio is used only as an IDE and as the source of the Windows SDK and headers.

Configure a standalone build

```
cmake -S . -B build -G Ninja \
  -DCMAKE_BUILD_TYPE=RelWithDebInfo \
  -DCMAKE_C_COMPILER=clang-cl \
  -DCMAKE_CXX_COMPILER=clang-cl
cmake --build build --parallel 8
ctest --test-dir build --output-on-failure
```

Use `RelWithDebInfo` for tests. A Debug build is valid but several GPU and video tests become an order of magnitude slower and can hit CI timeouts.

Engine options

Option	Default	Meaning
`FBY_BUILD_EDITOR`	`ON`	Dear ImGui editor library and application.
`FBY_BUILD_SANDBOX`	`ON`	Sandbox executable.
`FBY_BUILD_SERVER`	`ON`	Dedicated server executable.
`FBY_BUILD_LOBBY`	`ON`	Lobby service executable.
`FBY_BUILD_TESTS`	`OFF`	Build the CTest suite.
`FBY_BUILD_TOOLS`	`ON`	`fby_fempak`, `fby_aotc`, `fby_fontc`, `fby_scene_tool`.
`FBY_BUILD_PLATFORM_PROVIDERS`	`ON`	Steam/EGS/GOG provider shared libraries.

FBY_RHI_BACKEND	auto	auto, vulkan, metal, or null.
FBY_WITH_ASAN, FBY_WITH_UBSAN	OFF	Sanitizer instrumentation for the engine and tests.
FBY_ASAN_LEAKS	ON	LeakSanitizer at exit on Linux and macOS.
FBY_WITH_SCRIPT_VM	ON	FemboyScript VM for editor and development builds.
FBY_WITH_RNNOISE	ON	RNNoise neural denoise tier.
FBY_WITH_DLSS, FBY_WITH_FIDELITYFX, FBY_WITH_METALFX	OFF	Vendor upscaler backends.
FBY_DLSS_ROOT	third_party/dlss	Local NVIDIA DLSS SDK root; not tracked.
FBY_FIDELITYFX_ROOT	third_party/fidelityfx/Kits/FidelityFX	Local AMD FidelityFX SDK root; not tracked.
FBY_REQUIRE_VENDOR_SDKS	OFF	Missing vendor SDK becomes a configure error instead of a warning.
FBY_WITH_FSR1, FBY_WITH_FSR2, FBY_WITH_FRAME_GENERATION	OFF	Compatibility flags; the features are core and runtime-gated.
FBY_CONSOLE_APPS	OFF	Keep a console window for the editor and games.

The engine works without a Vulkan SDK: `FBY_RHI_BACKEND=null` and a null RHI build are supported for headless CI. If Vulkan headers are unavailable, the editor still builds against the minimal Khronos headers vendored by SDL.

Public CMake aliases

Link only PascalCase aliases. They are the supported build-tree API.

Alias	Purpose
FemboyEngine2D::Engine	Runtime library. Link this from a game.
FemboyEngine2D::Editor	Editor library and UI stack.
FemboyEngine2D::AssetCook	BC7/ASTC texture cooker.
FemboyEngine2D::AssetLib	Pack and localization compiler.
FemboyEngine2D::ShaderCompilerCore	In-process GLSL/SPIR-V compiler.

Internal targets (`fby_engine`, `fby_editor`, `fby_fempak`, `fby_aotc`) and executable names are implementation details. They are not renamed for style and must not be used as a public contract. There are no `FemboyEngine::*` or `FemboyEngine2D::Runtime` compatibility aliases.

Consume the engine with add_subdirectory

```
cmake_minimum_required(VERSION 3.28)
project(MyGame LANGUAGES CXX)

set(FBY_ENGINE_ROOT "${CMAKE_CURRENT_SOURCE_DIR}/../.." CACHE PATH
"FemboyEngine2D root")
set(FBY_BUILD_EDITOR OFF CACHE BOOL "" FORCE)
set(FBY_BUILD_SANDBOX OFF CACHE BOOL "" FORCE)
set(FBY_BUILD_TESTS OFF CACHE BOOL "" FORCE)
set(FBY_BUILD_TOOLS OFF CACHE BOOL "" FORCE)
set(FBY_BUILD_PLATFORM_PROVIDERS OFF CACHE BOOL "" FORCE)

add_subdirectory("${FBY_ENGINE_ROOT}" "${CMAKE_BINARY_DIR}/_fby_engine"
EXCLUDE_FROM_ALL)

add_executable(MyGame GameMain.cpp)
target_link_libraries(MyGame PRIVATE FemboyEngine2D::Engine)
target_compile_definitions(MyGame PRIVATE
    FBY_PROJECT_ASSETS="${CMAKE_CURRENT_SOURCE_DIR}/assets")
```

The binary directory must be unique. Do not add the engine as the same binary directory as the game; CMake will reject the target names.

`FBY_PROJECT_ASSETS` is a compile definition, not an engine cache variable. The generated `GameMain.cpp` uses it to mount the project asset directory.

Generated Clean Slate project

`CreateProject()` writes:

```
MyGame/
  CMakeLists.txt
  project.femproj
  GameMain.cpp
  assets/
```

The generated CMakeLists disables editor, sandbox, tests, tools and storefront providers. It links only `FemboyEngine2D::Engine` and defines `FBY_PROJECT_ASSETS`. Open it with:

```
cmake -S MyGame -B MyGame/build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo
cmake --build MyGame/build --parallel 8
```

The engine root must be passed by the project because the default project

location is `<engine>/Projects/<name>`. The generated CMake also accepts the `FBY_ENGINE_ROOT` environment variable; this keeps the project portable when it is moved to another machine or operating system:

```
export FBY_ENGINE_ROOT=/path/to/FemboyEngine2D
cmake -S MyGame -B MyGame/build
```

No absolute engine path is stored in `project.femproj`. Asset paths inside the project are project-relative; the generated CMake computes the absolute `FBY_PROJECT_ASSETS` path at configure time, so moving the project requires a reconfigure/rebuild, not a data migration.

An explicit cache value still wins:

```
cmake -S MyGame -B MyGame/build -DFBY_ENGINE_ROOT=/path/to/FemboyEngine2D
```

Build products

Target	Output
`fby_editor_app`	`build/bin/fby_editor`
`fby_sandbox`	`build/bin/fby_sandbox`
`fby_fempak`	`build/bin/fby_fempak`
`fby_aotc`	`build/bin/fby_aotc`
`fby_fontc`	`build/bin/fby_fontc`
`fby_scene_tool`	`build/bin/fby_scene_tool`

Tests and style gate

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo
-DFBY_BUILD_TESTS=ON
cmake --build build --parallel 8
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

The CTest suite builds standalone executables with no test framework. They run headless and can be launched individually from `build/bin`.

RHI selection

```
cmake -S . -B build -DFBY_RHI_BACKEND=vulkan
cmake -S . -B build -DFBY_RHI_BACKEND=null
```

`null` is a real backend with deterministic frame accounting, not a stub. It is used by CI and by the generated project when no GPU is present. `auto` chooses Metal on macOS and Vulkan on Windows and Linux when headers are available.

Adding a target

1. Add the source files to `Source/Engine/CMakeLists.txt` or the owning subsystem CMakeLists.
2. Link `FemboyEngine2D::Engine`, not `fby_engine`.
3. Add public include directories as `$<BUILD_INTERFACE:...>`.
4. Add a test through `fby_add_engine_test()` in `Source/Tests/CMakeLists.txt`.
5. Run the style gate and CTest.

Troubleshooting

- `cmake/EmbedShaders.cmake` not found: the engine was included from a game and `CMAKE_SOURCE_DIR` points at the game. Use the shipped `add_subdirectory` pattern; the engine derives its own root from `CMAKE_CURRENT_SOURCE_DIR`.
- `vulkan/vulkan.h` missing: configure with `-DFBY_RHI_BACKEND=null`, or install Vulkan-Headers. The editor has an SDL Khronos-header fallback.
- Legacy alias errors: replace `FemboyEngine::*` and `FemboyEngine2D::Runtime` with `FemboyEngine2D::Engine`.
- Tests time out: use `RelWithDebInfo`, not `Debug`.
- C++26 errors: update the compiler. C++23 is not supported.